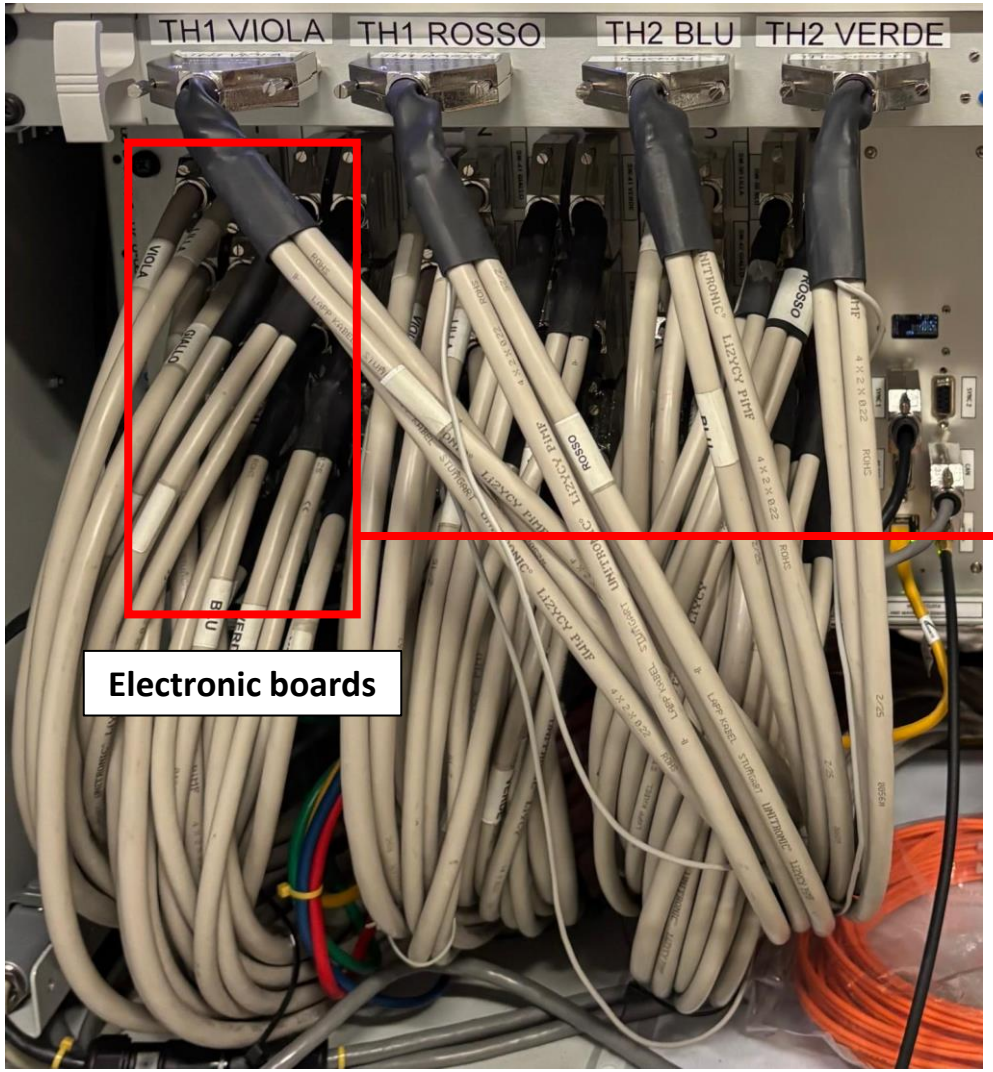
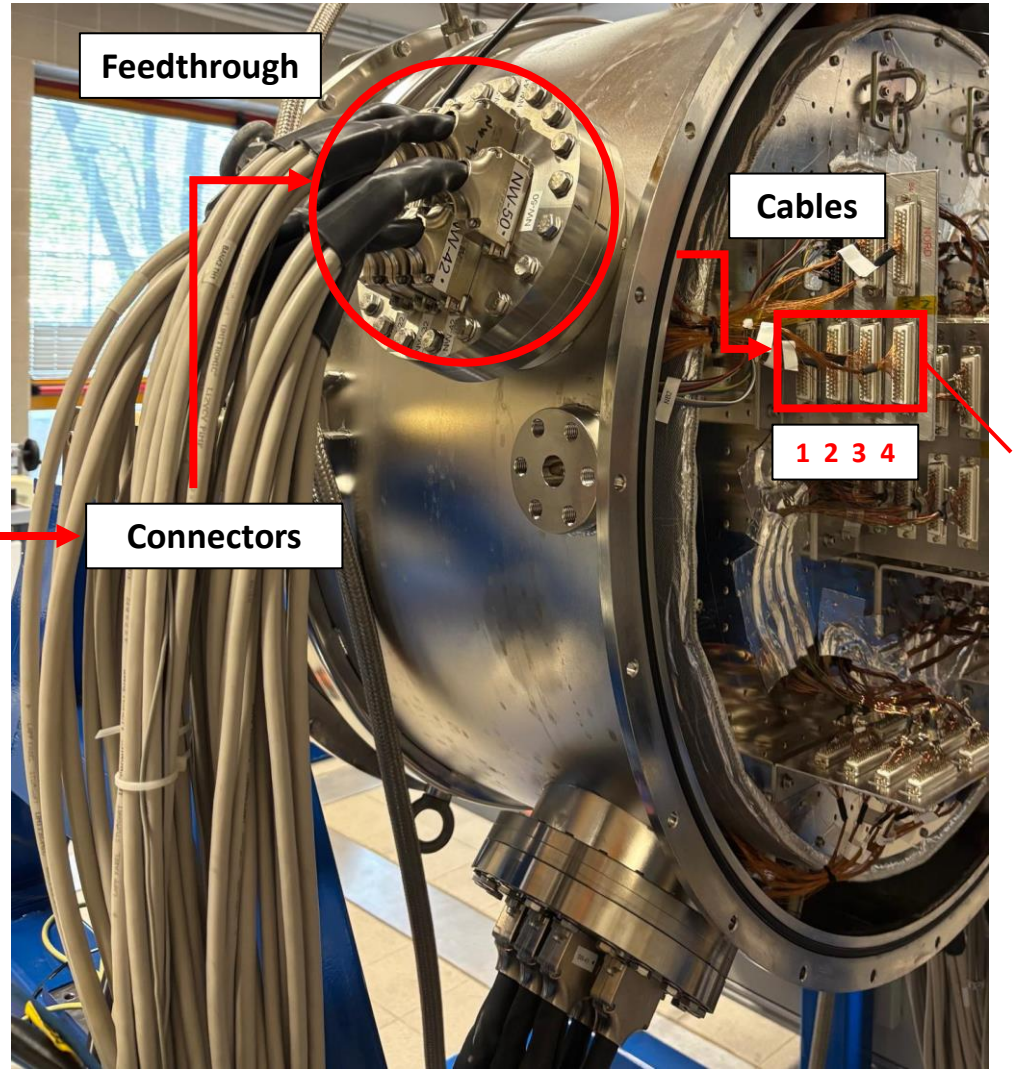
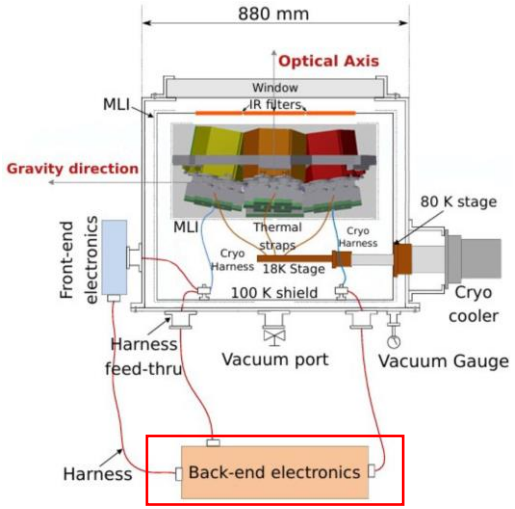
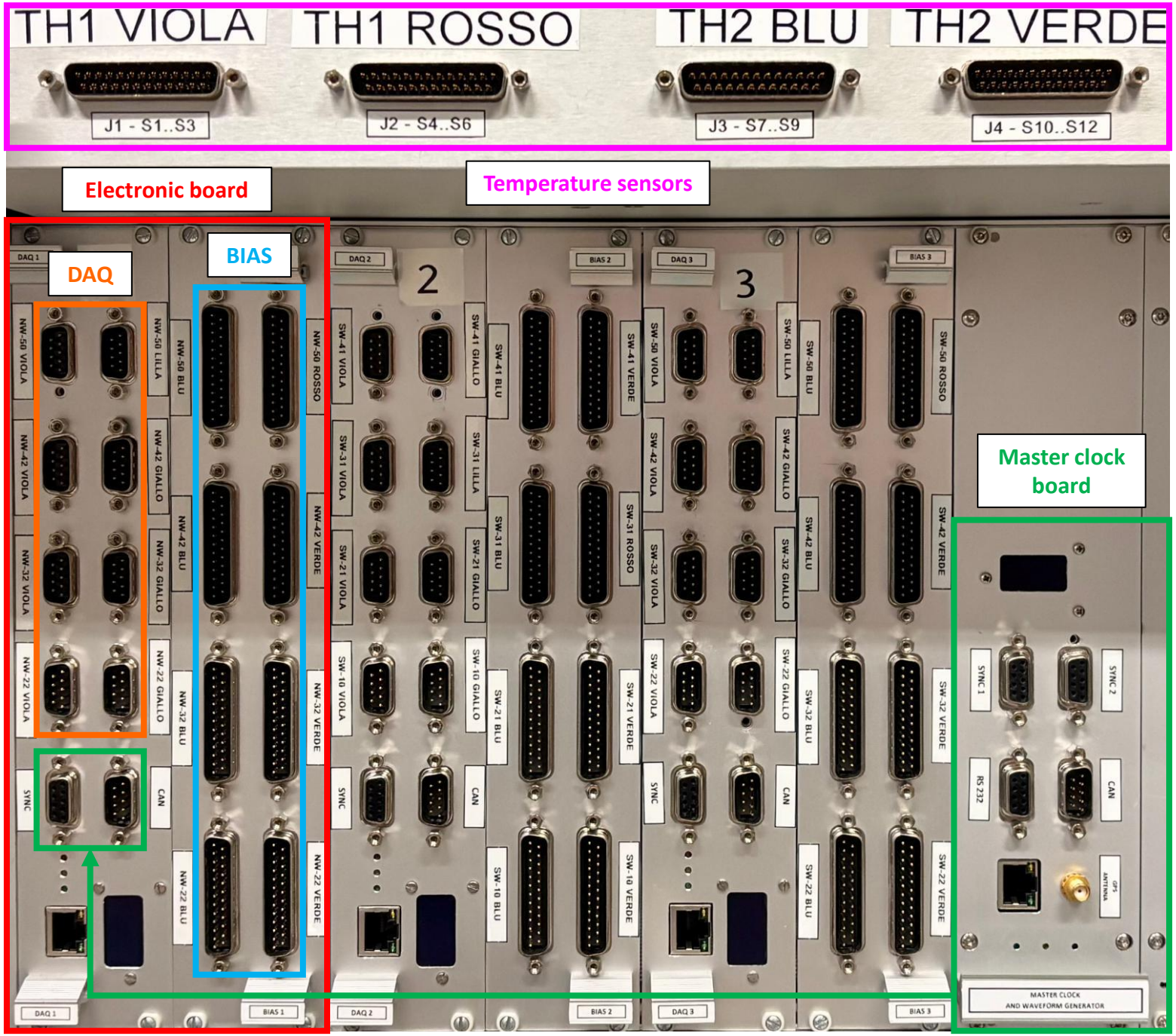
LSPE-Strip cryostat

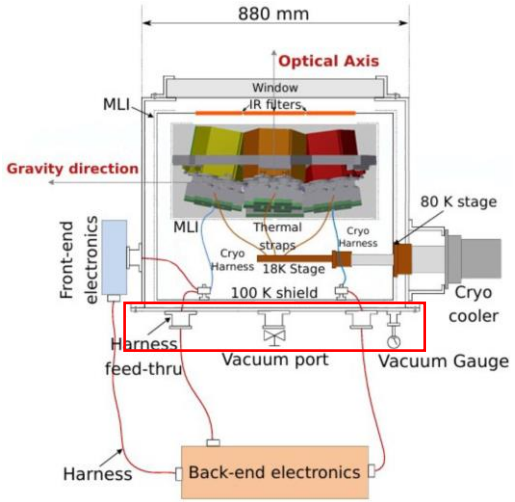


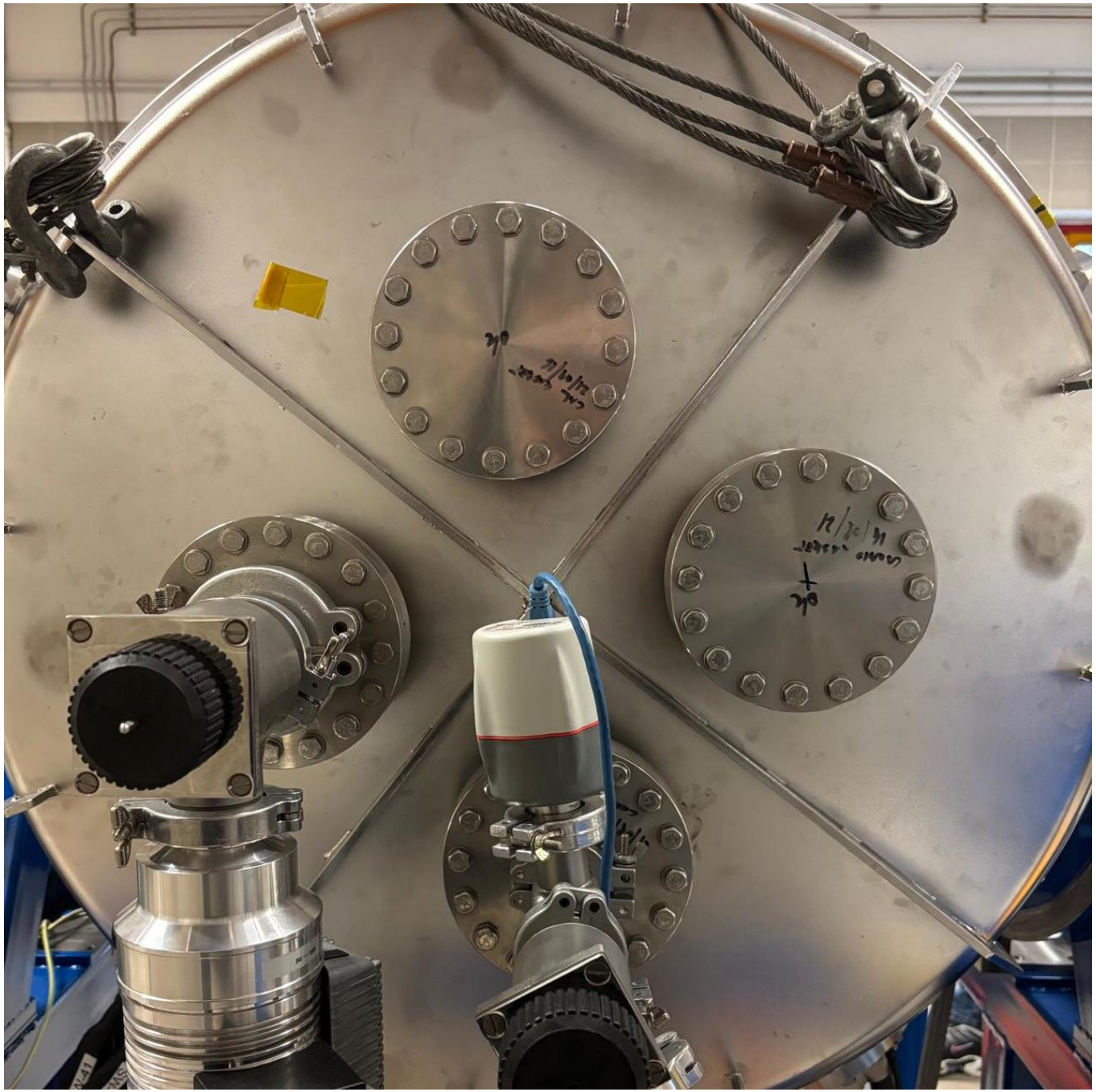


50-pin connectors, each one serving a pair of polarimeters


880 mm
Optical Axis
Window
IR filters
MLI
Gravity direction
80 K stage
Front-end electronics
MLI
Thermal straps
Cryo Harness
Cryo Harness
18K Stage
100 K shield
Cryo cooler
Harness feed-thru
Vacuum port
Vacuum Gauge
Harness
Back-end electronics

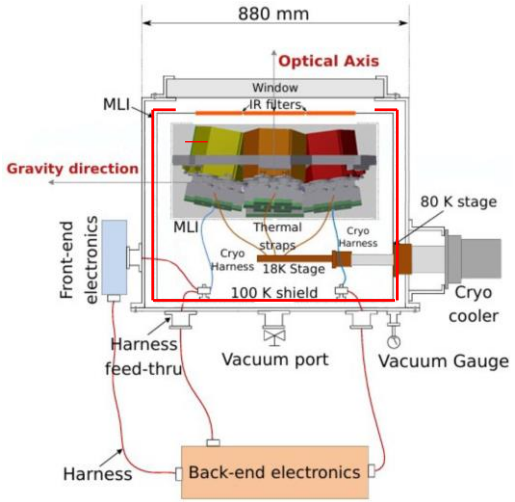

TH1 VIOLA
TH1 ROSSO
TH2 BLU
TH2 VERDE
J1 - S1..S3
J2 - S4..S6
J3 - S7..S9
J4 - S10..S12
Electronic board
Temperature sensors
DAQ
BIAS
Master clock board
MASTER CLOCK AND WAVEFORM GENERATOR

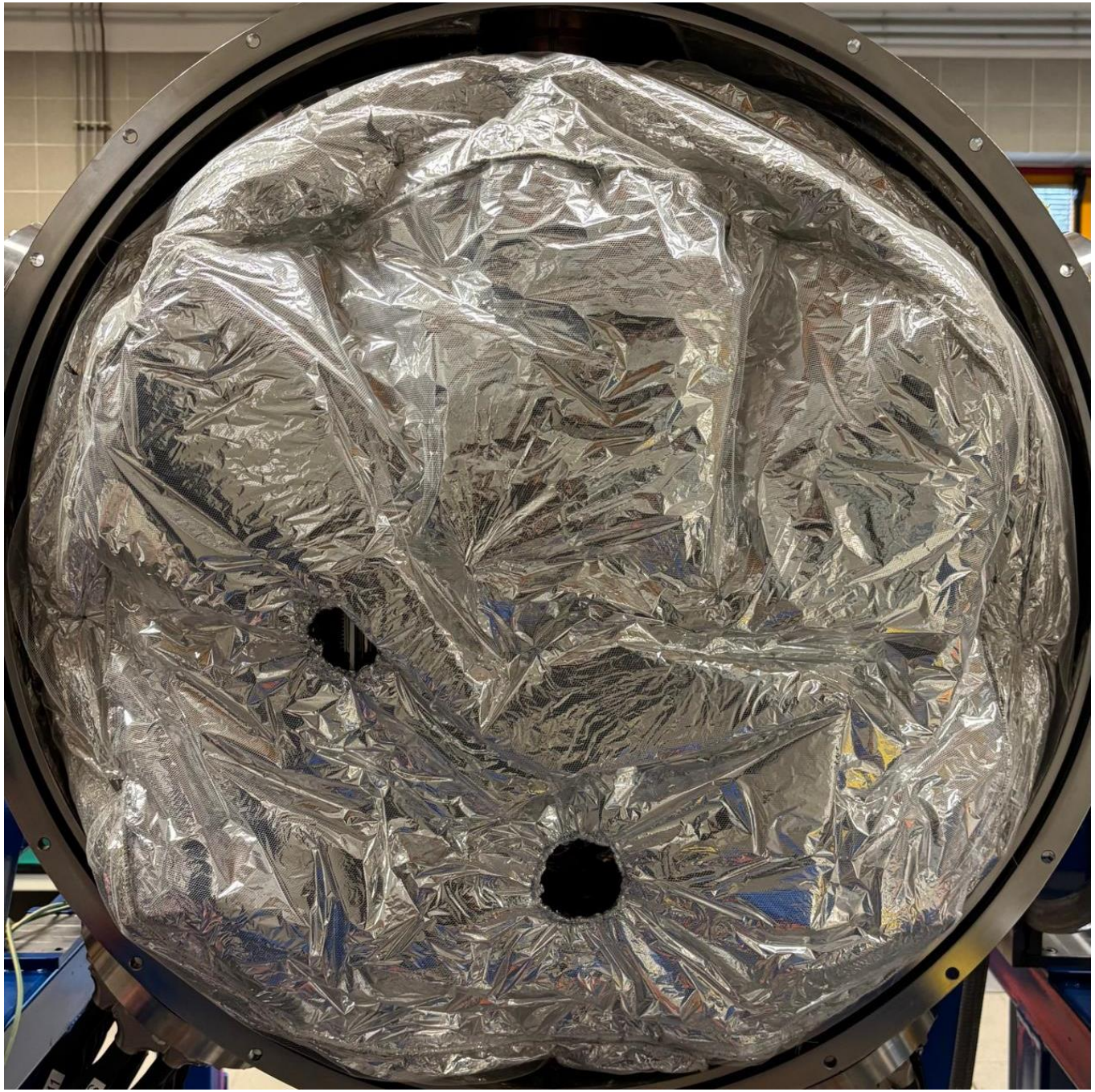




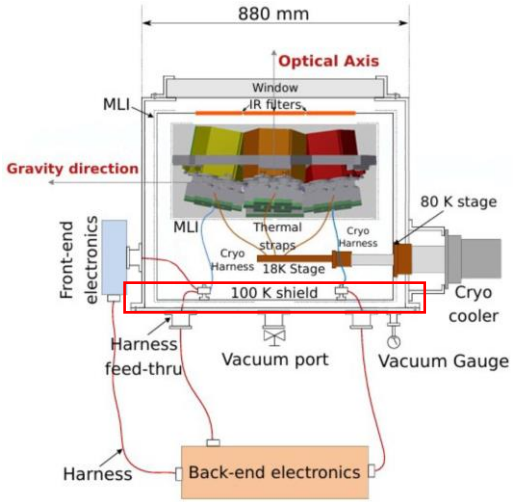
Rotary pump provides a low vacuum pressure ($< 10^{-2}$ mbar)

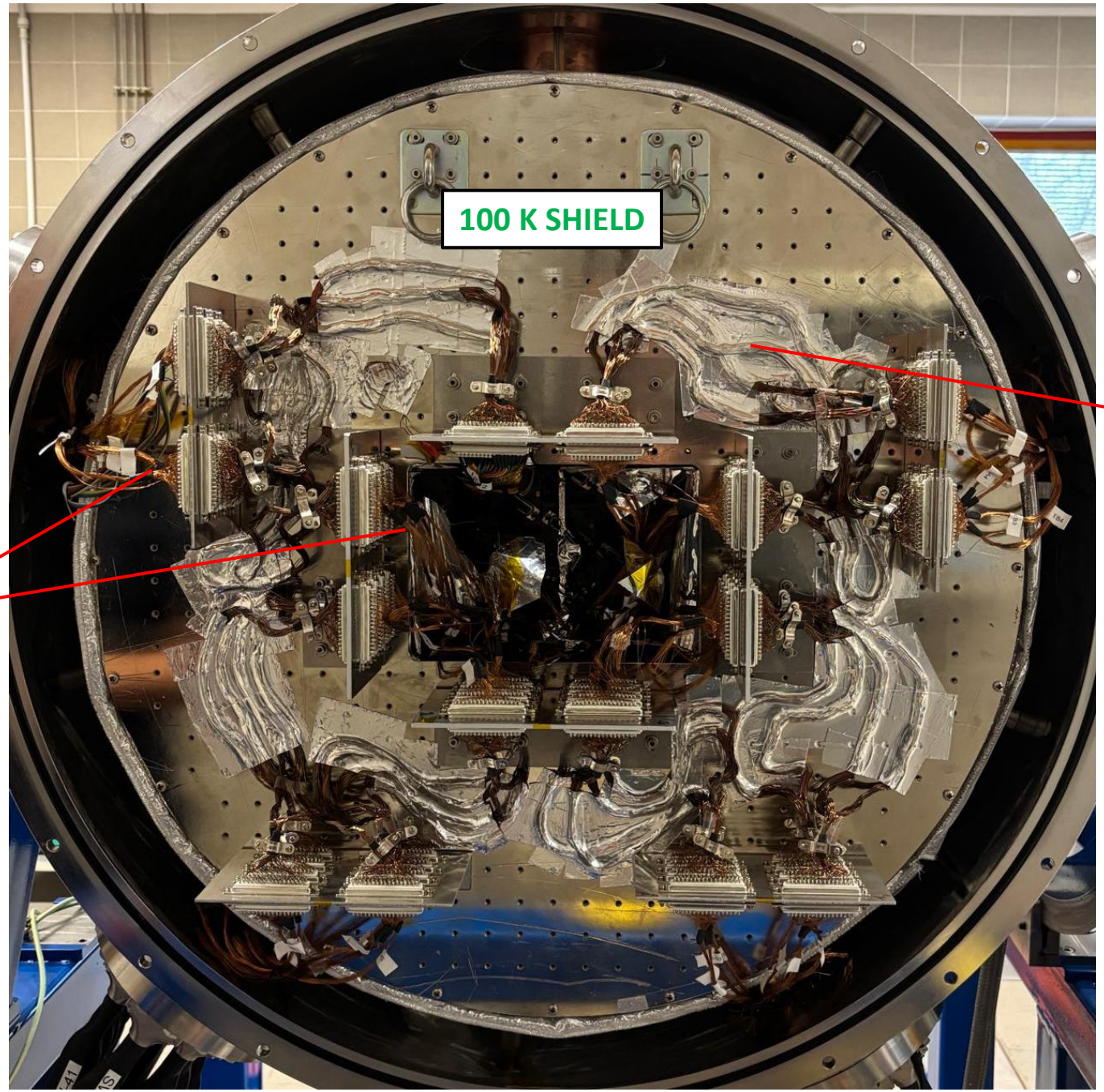
Turbo-molecular pump provides a high vacuum pressure ($< 10^{-6}$ mbar)





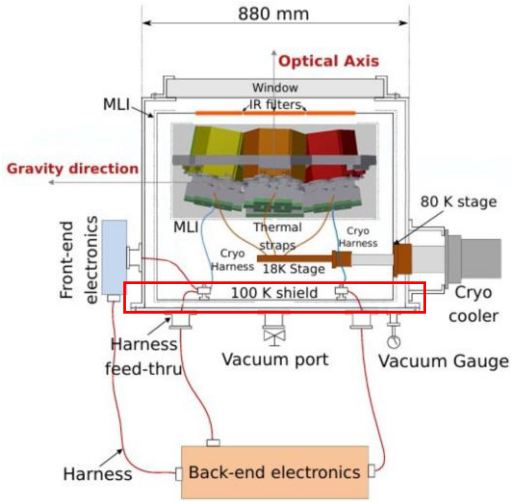
Multi-layer insulation (MLI), a thermal blanket made of alternating layers of aluminized plastic films, that reduces radiative heat transfer





Copper cables thermalize the wires at 100 K

Manganine cables connect the external shell (300 K) with the 100 K thermal shield, and then connect with the internal stage (20 K)


880 mm
Optical Axis
Window
IR filters
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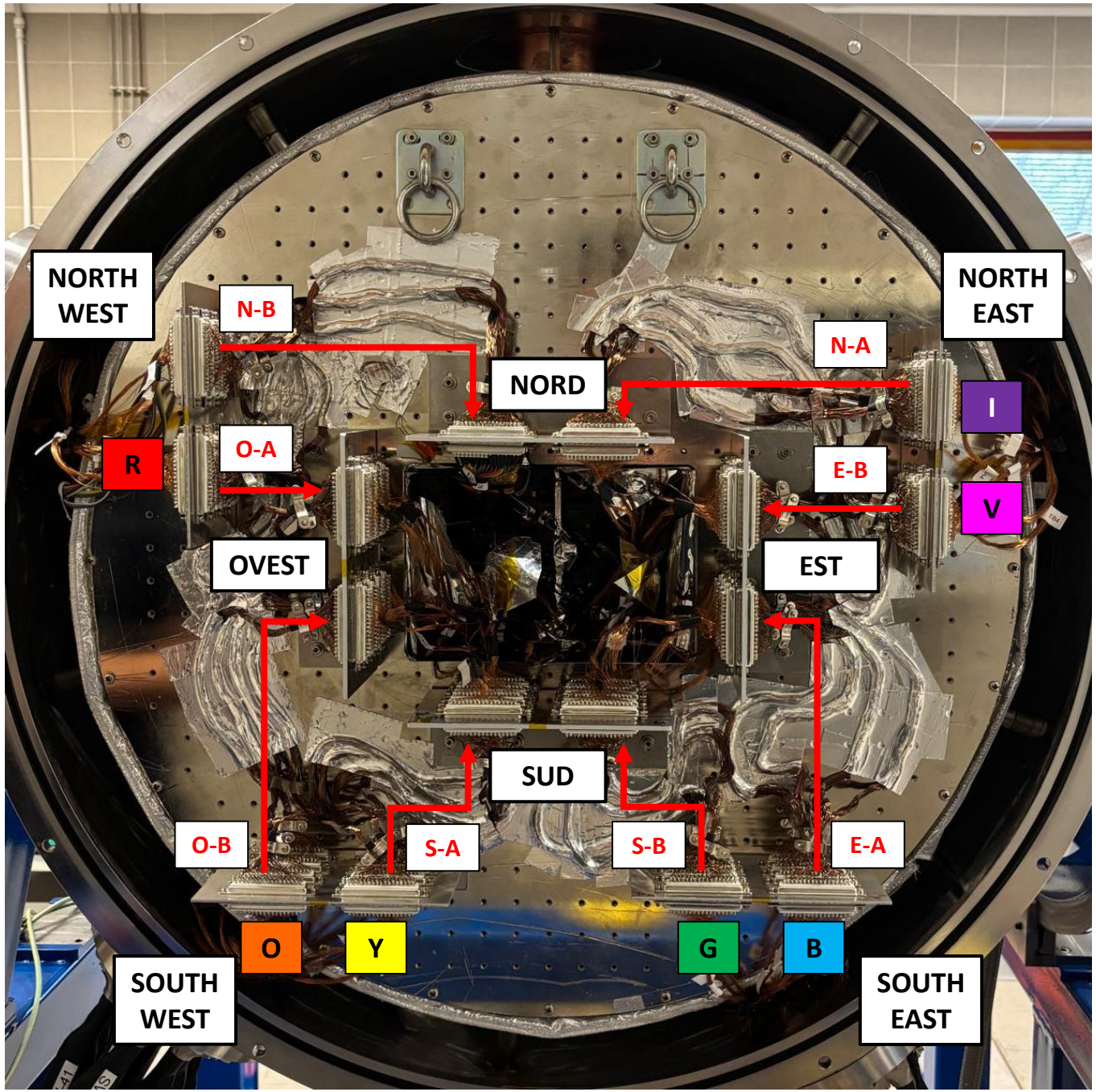
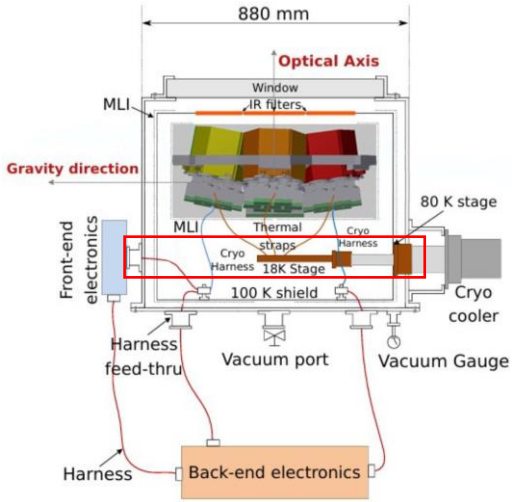
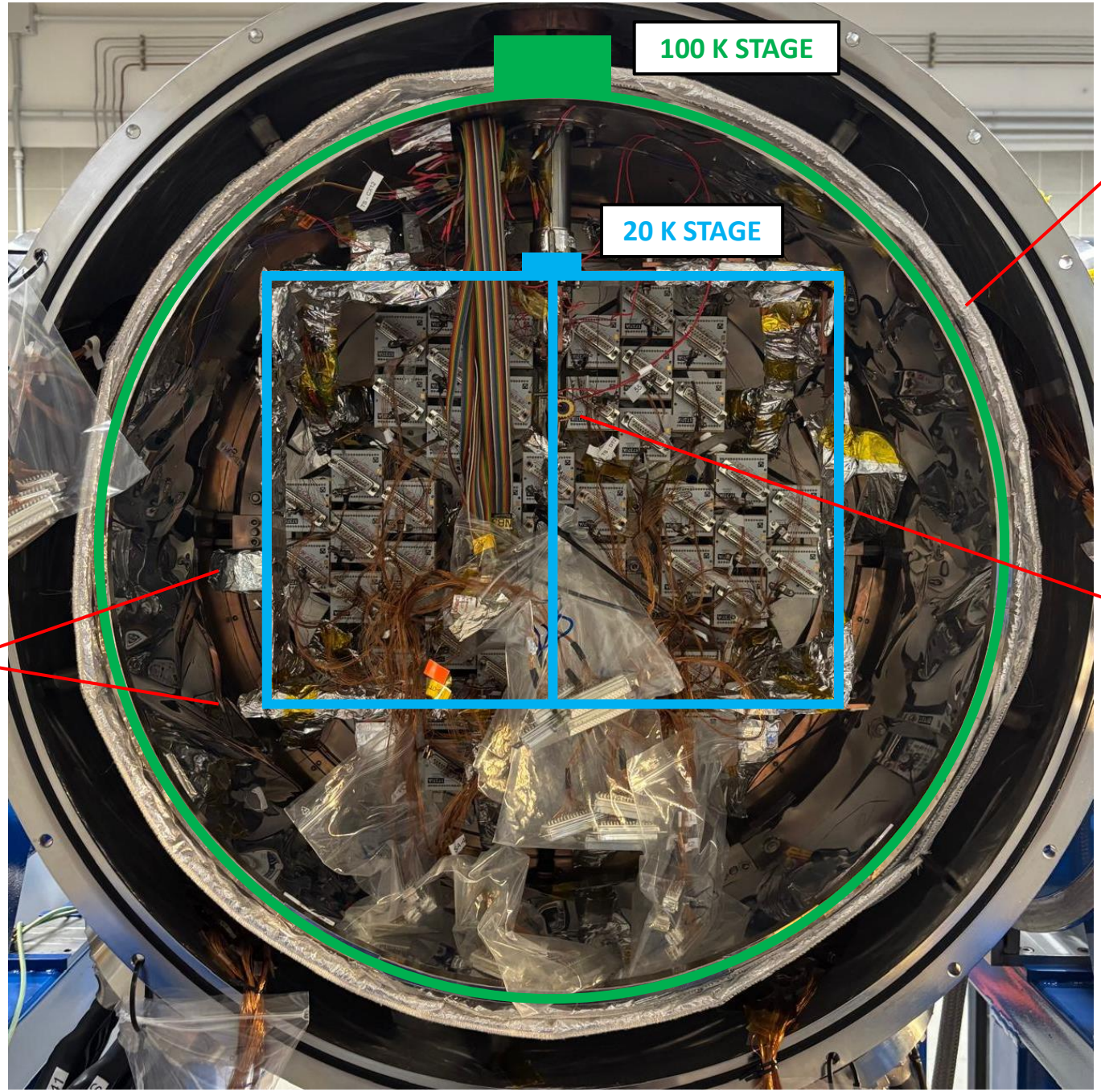
Gravity direction
80 K stage
MLI
Thermal straps
Cryo Harness
Cryo Harness
18K Stage
Front-end electronics
100 K shield
Cryo cooler
Harness feed-thru
Vacuum port
Vacuum Gauge
Harness
Back-end electronics


NORTH WEST
NORTH EAST
N-B
N-A
NORD
I
R
O-A
E-B
V
OVEST
EST
SUD
O-B
S-A
S-B
E-A
O
Y
G
B
SOUTH WEST
SOUTH EAST

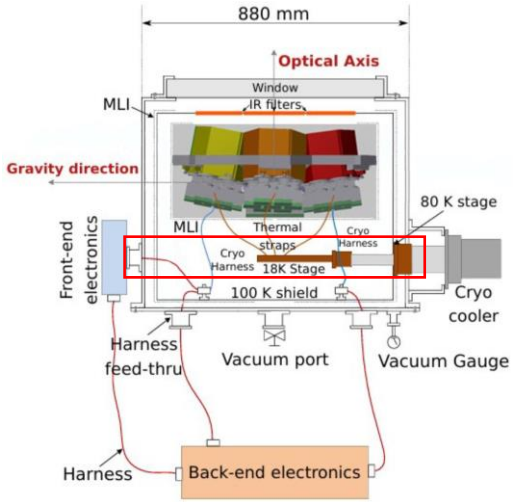


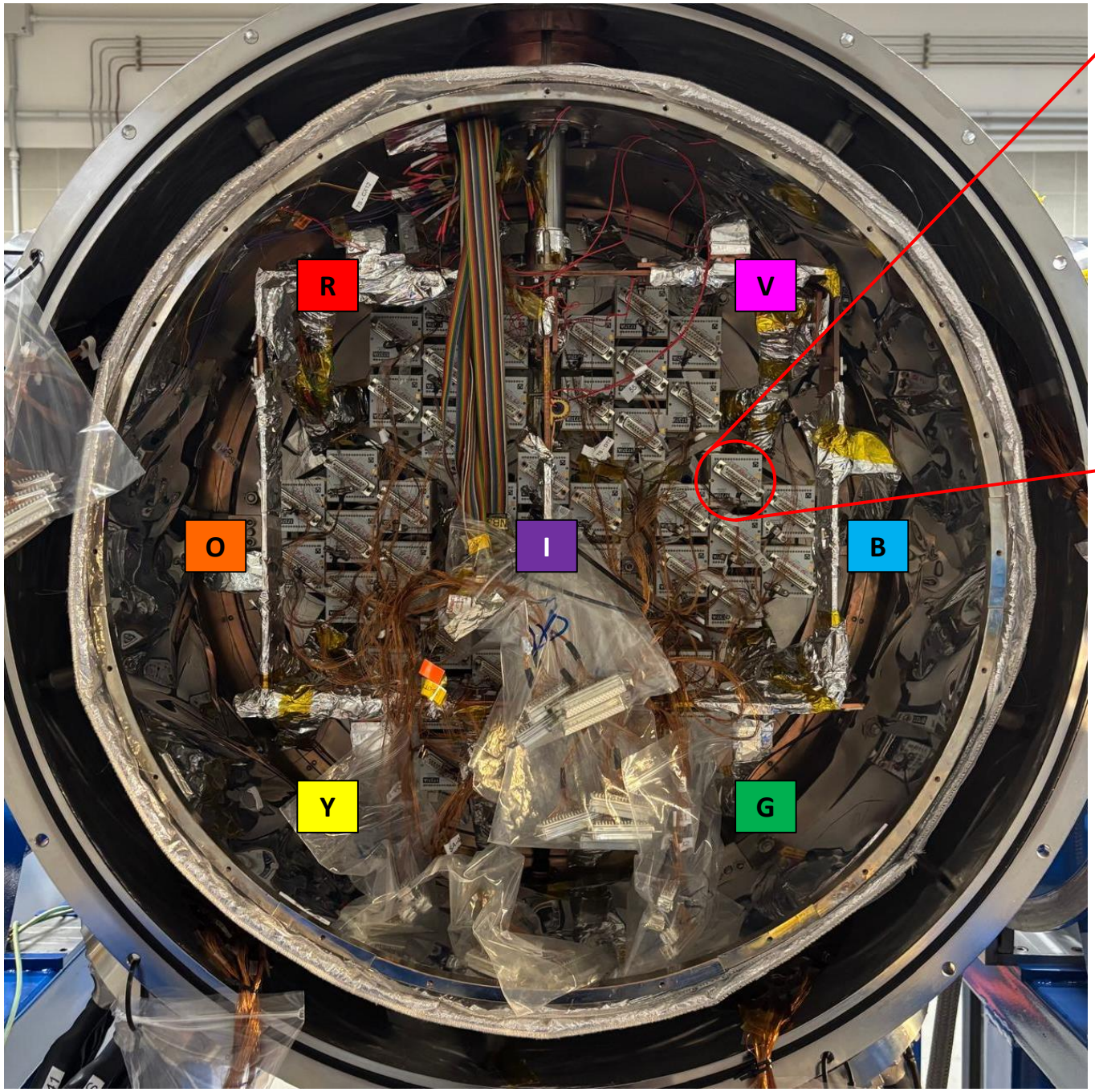


Radiative shield surrounding the focal plane, insulated by MLI

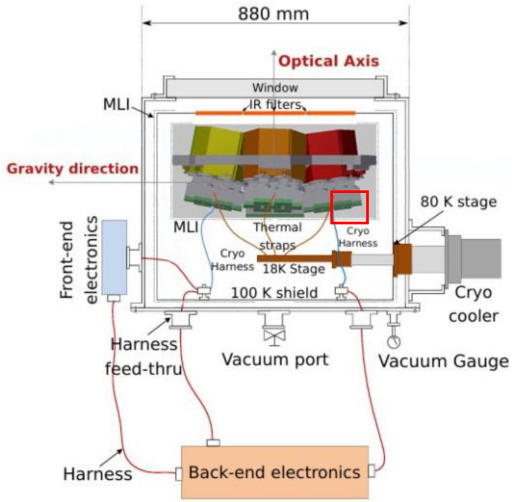
PID controller, composed by a thermal sensor and an heater that keeps the temperature stable in the cryostat

"H" frame thermally connected to the focal plane by *thermal straps*, allowing the polarimeters and feedhorns to be cooled to 20 K

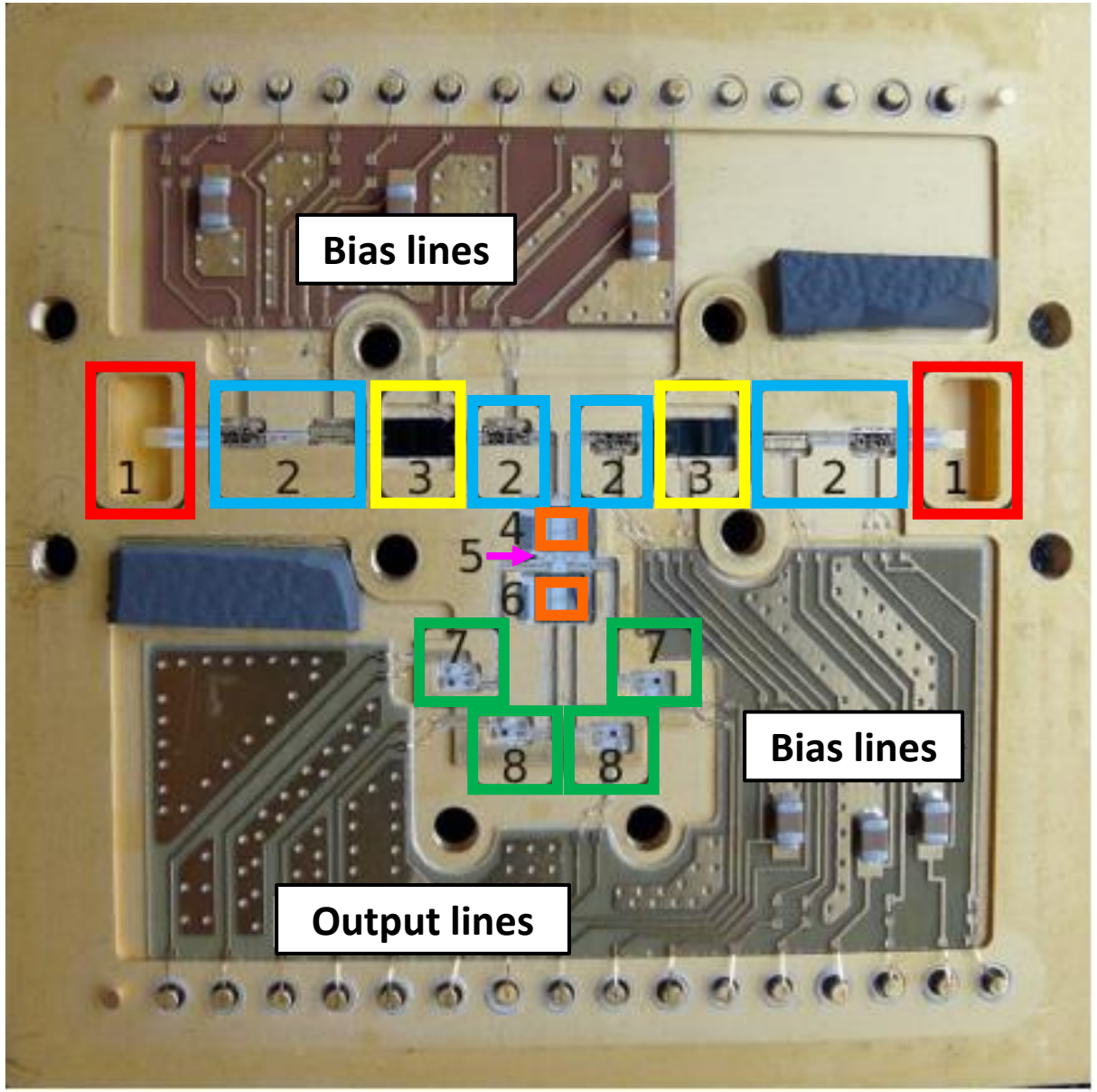




25-pin connector linked to the polarimeter

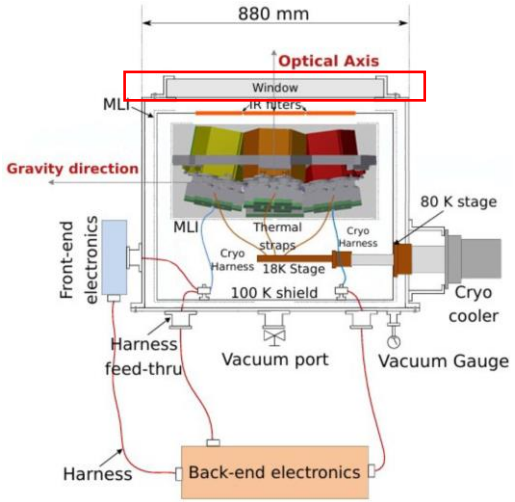


~ 5 cm

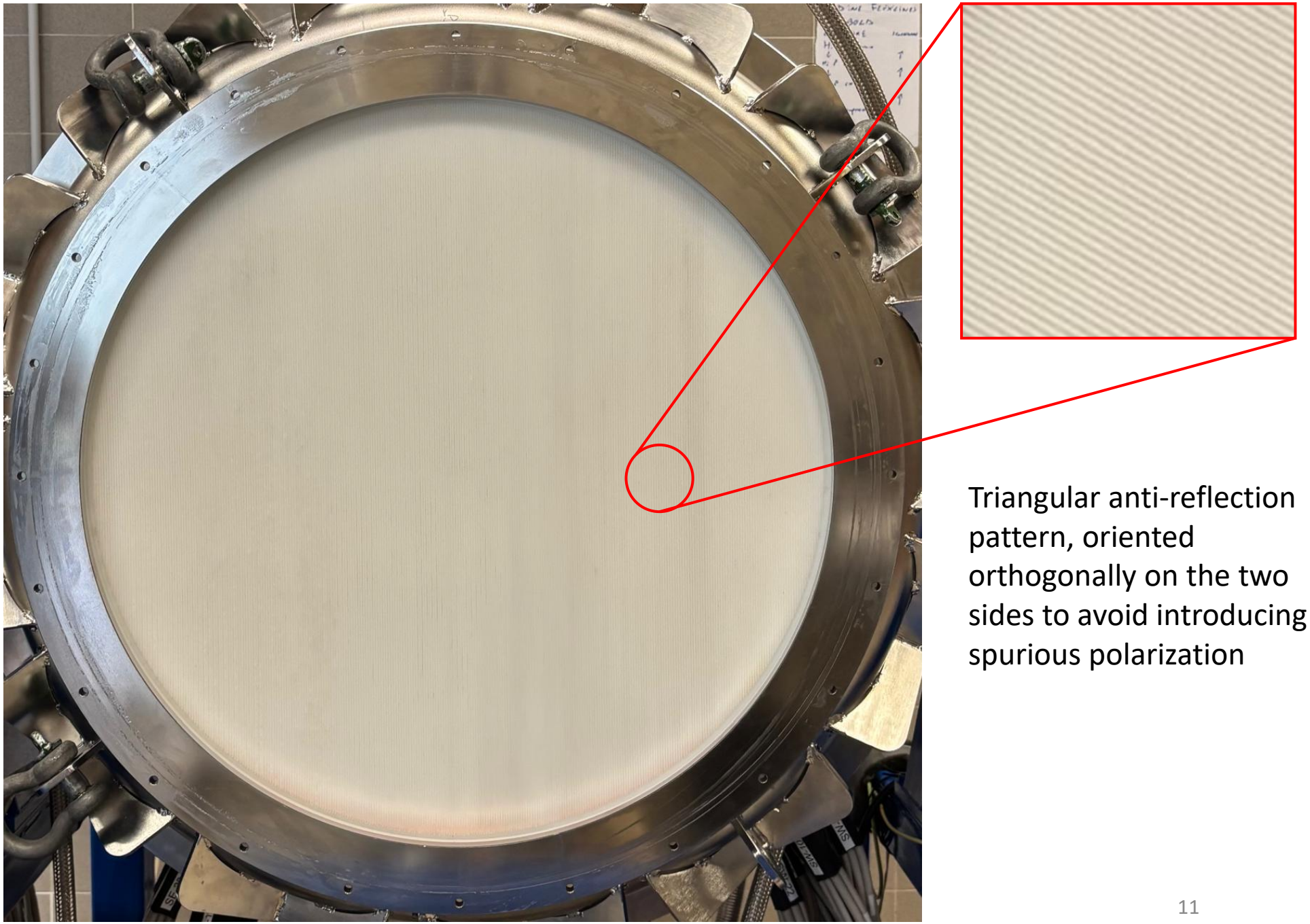


Scheme of the polarimeter:

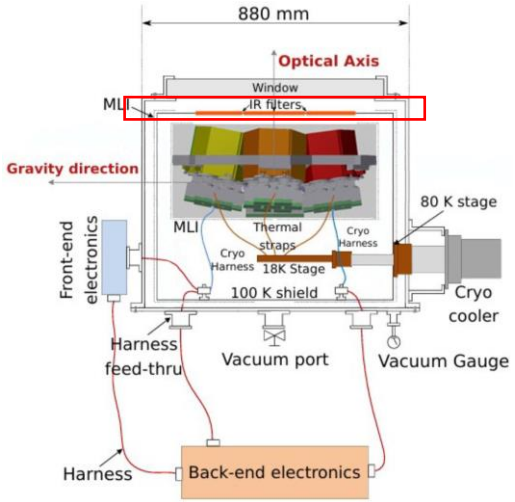
1. Inputs
2. Low-noise amplifiers
3. Phase switches
4. 180° hybrid
5. Power splitters
6. 90° hybrid
7. Q-diodes
8. U-diodes

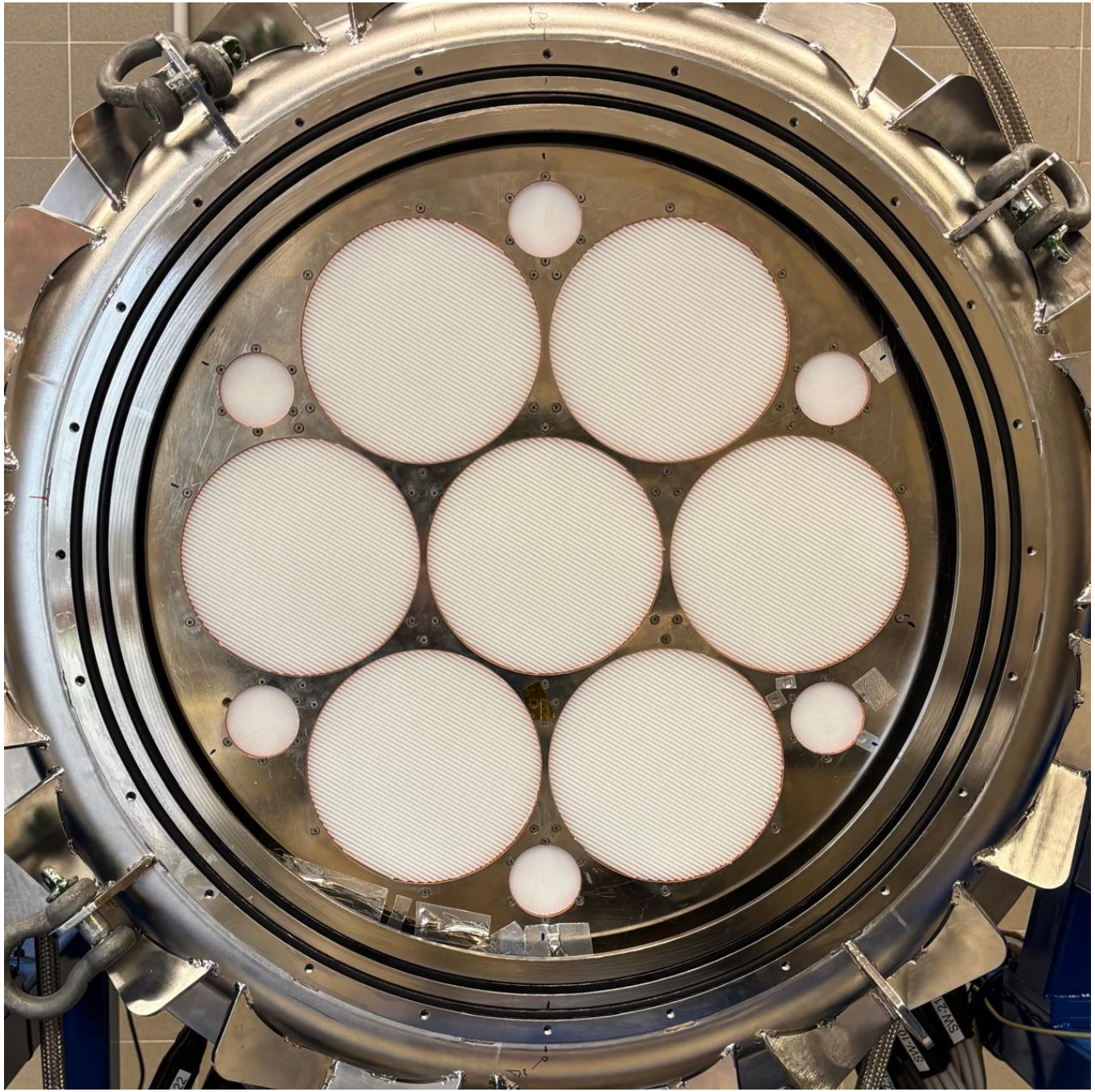


Vacuum window of ultra high molecular weight polietilene (UHMWPE), which closes the side of the cryostat facing the telescope and transmits the incoming microwave radiation, while preserving the vacuum inside the chamber

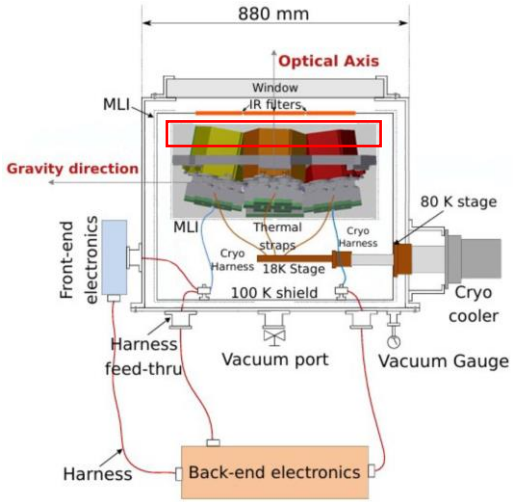


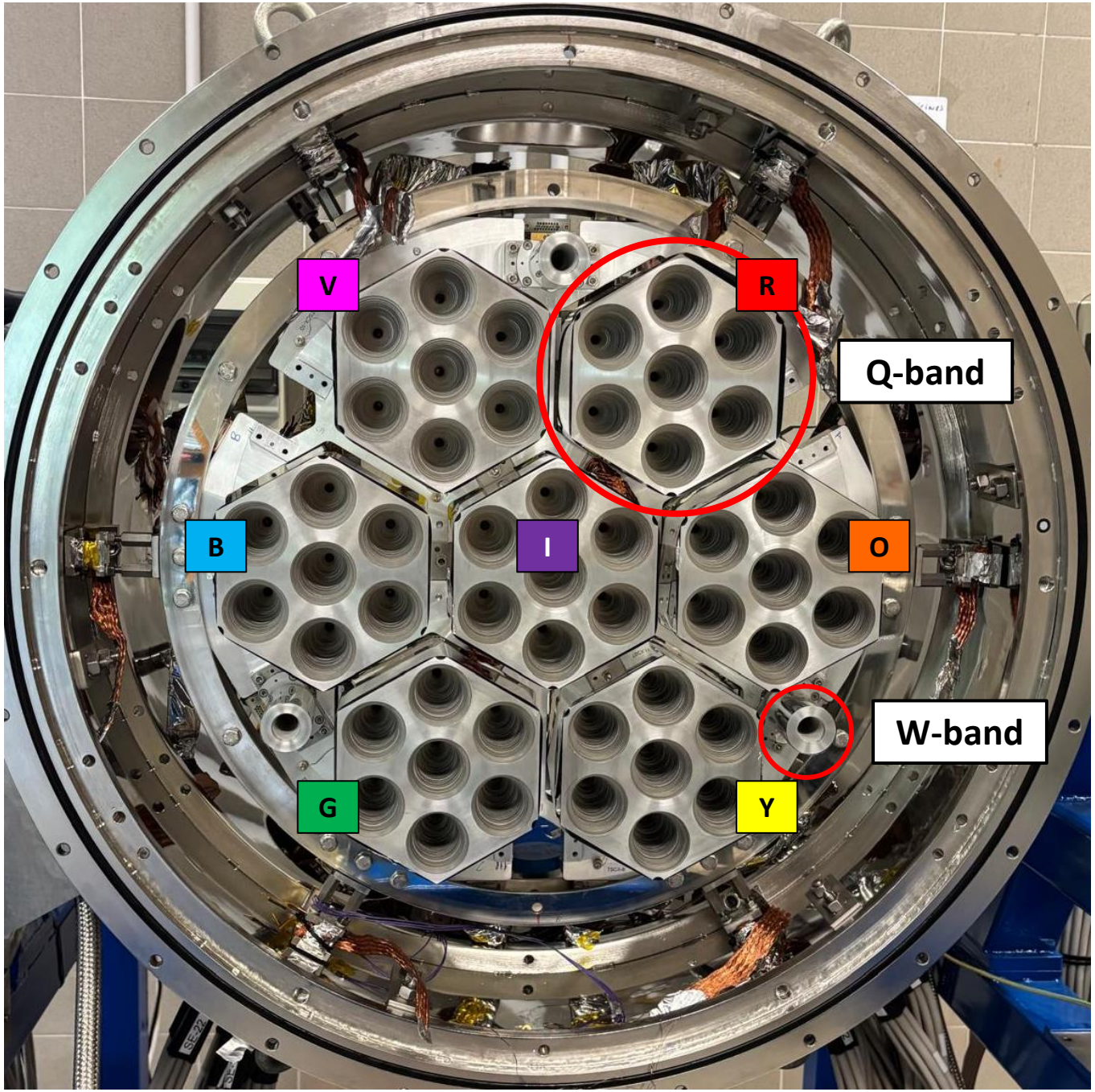
Triangular anti-reflection pattern, oriented orthogonally on the two sides to avoid introducing spurious polarization





Series of *IR filter* of polytetrafluoroethylene in front of the horns and thermalized at 100 K, in order to absorb and re-emit the IR radiation from the outside at a lower temperature, reducing the thermal load on the coldest stage





Corrugated feedhorns array at 43 GHz (Q-band) and 95 GHz (W-band), made of aluminum with a dual profile